Carlos Armando Aguila, MS

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Selected Professional Experiences

Center for Neuroengineering and Therapeutics, University of Pennsylvania **Philadelphia, PA**
Graduate Research Fellow, Advisor: Erin C. Conrad, MD & Brian Litt, MD 2022 - Present

- Conducted time-series analysis (calculate spike rates, multidien spike cycles, networks, etc.) of intracranial electroencephalography (EEG) data across a multi-center cohort of patients.
- Developed machine learning models to integrate interictal and ictal features to predict mesial temporal lobe epilepsy.
- Collaborated with graduate students to characterize ictal intracranial EEG using a novel unsupervised deep learning network to cluster similar seizure dynamics across patients.

Neural Memos (*https://neuralmemos.substack.com/*) **Philadelphia, PA**
Founder & Author 2025 - Present

- Authoring a series of technical deep dives analyzing the commercial viability, technological hurdles, and key innovators within the Brain-Computer Interfaces sector.
- Translating neurotechnology concepts into strategic insights for venture investors & industry stakeholders.

Penn Biotech Group, University of Pennsylvania **Philadelphia, PA**
Consultant, Supervisor: Ana Defendini, PhD 2022 - Present

- Carried out comprehensive analyses of academic technologies to assess commercial viability and facilitate commercialization of promising ones.
- Conducted in-depth market landscape research, outlining where a technology might fit within the commercialization pipeline and the industry environment
- Drafted invention assessments, detailing a foundational understanding of a technology and commercializing potential.

Physiomimetic Microsystems Laboratory (PML), University of Miami **Miami, FL**
Research Assistant, Advisor: Ashutosh Agarwal, PhD 2019 - 2021

- Designed and optimized organ-on-chip platforms to model shear stress, oxygen diffusion, and cancer extravasation.

Department of Psychiatry and Behavioral Sciences, University of Miami **Miami, FL**
Research Assistant, Advisor: Deborah Weiss, PhD Summer 2017

- Automated analysis and translation of clinical interviews from Zambian HIV prevention clinics to evaluate program effectiveness.

Siemens South Miami **Miami, FL**
Intern, Supervisor: Guillermo Padron Summer 2017

- Shadowed engineers troubleshooting, repairing, and programming Siemens medical imaging equipment.

Selected Leadership Experiences

Nucleate, Philadelphia Chapter **Philadelphia, PA**
Co-Head Technology and Partnership 2025 - Present

- Engage directly with Philadelphia-area technology transfer offices and inventors to recruit early-stage technologies for Nucleate Philadelphia's 2025 Activator cohort.
- Cultivate and manage a regional partnership portfolio of VCs, law firms, and life science companies to secure financial/in-kind support, mentorship, and resources for Philadelphia's academic-led biotech ventures.

Selected Skills

Languages: Native in Spanish, fluent in English
Programming: Highly Proficient: Python, MATLAB Comfortable with: HTML, CSS, R Familiar with: C, C++, Java
Machine Learning and Data Analysis: Clustering, Unsupervised/Semi-Supervised/Supervised Learning, Deep Learning (Artificial Neural-Networks), Time Series Analysis, Image Analysis
Signal Processing & Domain: EEG/iEEG analysis, biomedical time-series, feature engineering, automated event detection

Education

University of Pennsylvania Philadelphia, PA
PhD Candidate in Bioengineering, Expected: Fall 2026 2021-Present
Dissertation: Mapping dynamic epileptic networks using spontaneous and evoked interictal EEG
Selected Honors: Fontaine Fellowship, NSF Graduate Research Fellowship Program (GRFP)

University of Miami **Miami, FL**
Bachelor of Science in Biomedical Engineering *2016 - 2020*
Selected Honors: Magna Cum Laude, Alpha Eta Mu Beta Award, Ronald A. Hammond Scholarship, Florida-Georgia Louis Stokes Alliance for Minority Participation (FGLSAMP)